

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Smart Classroom

Requirements Baseline – Version 1.0

Document Information

Item	Details
project	SmartClassroom
Organization	University College Of Jaffna
Document	Software Requirements Specification
Version	1.0
Baseline	Requirements Baseline v1.0
Development Approach	Incremental Development with Controlled Change Management
Technology	React, Fastapi, MySQL
Version Control	Git and GitHub
API Testing	Postman
Status	Proposed for Review
Prepared By	Student Project
Date	31 st August 2026

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for Version 1.0 of the SmartClassroom system for University College Of Jaffna. The document establishes the agreed requirements that will be used as the reference for system analysis, system design, database design, software development, testing, and release activities.

1.2 Project Background

The proposed system is a web-based classroom application designed to digitize and streamline the assignment submission, tracking, grading, and communication process between students, teachers, and administrators. It provides role-based access, assignment uploads, submission tracking, grading, and automated email notifications.

2. Problem Statement

Traditional methods of managing assignments through email or physical submissions make it difficult to track who has completed or missed work. Teachers face challenges in managing submissions and grading manually, while students may miss deadlines because reminders are not automated. A centralized classroom system is therefore required to improve assignment tracking, grading, communication, and administrative control.

- Assignment information and submissions may be distributed across different channels.
- Teachers may spend significant time identifying completed and pending submissions.
- Manual grading and result communication can be time-consuming.
- Students may miss assignment deadlines without automated reminders.
- A centralized dashboard can improve visibility and coordination.

3. Project Objectives

1. To develop a secure multi-role authentication system for Students, Teachers, and Administrators.
2. To provide a file upload and tracking mechanism for student assignments.
3. To enable teachers to review submissions, track completed and pending students, and assign marks.
4. To automate email notifications for assignment openings, deadlines, and results.
5. To give administrators control over user management, including adding teachers and students.
6. To provide a responsive and user-friendly web application.
7. To improve the accuracy, visibility, and efficiency of assignment management.

4. Project Scope

4.1 In Scope for Version 1.0

- Role-based login for Students, Teachers, and Administrators.
- Student assignment upload and submission tracking.
- Teacher assignment creation and management.
- Teacher visibility of submitted and pending students.
- Assignment grading and recording of marks.
- Automated email notifications for assignment opening, deadlines, and results.
- Administrator management of teachers and students.
- Responsive web interface.
- REST API communication between frontend and backend.
- MySQL database storage.

4.2 Out of Scope for Version 1.0

- Features not identified in the supplied project proposal as core Version 1.0 functionality.
- Advanced functionality beyond assignment management, submission tracking, grading, notifications, and basic user administration.

5. Stakeholders

5.1 Stakeholder Register

Stakeholder	Role	Interest / Responsibility
Administrator	System User	Manage teachers and students and oversee system information
Teacher	Academic User	Create/manage assignments, track submissions, and grade work
Student	End User	View assignments, upload submissions, and receive updates
University / Management	Management Stakeholder	Oversee assignment workflow and academic progress
IT / Project Team	Technical Stakeholder	Analyze, design, develop, test, and maintain the system
Lecturer / Supervisor	Academic Supervisor	Monitor project progress and requirements practices

6. User Roles

6.1 Administrator

The administrator is responsible for managing system users and overall system administration.

- Log in and log out.
- Add and manage teachers.
- Add and manage students.
- Oversee overall system progress and user information.

6.2 Teacher

Teachers manage assignments and evaluate student submissions.

- Log in and log out.
- Create and manage assignments.
- Set assignment opening and deadline information.
- View submission statistics.
- Identify completed and pending students.
- Review submitted assignment files.
- Assign and record marks.

6.3 Student

Students use the system to access assignments and submit their work.

- Log in and log out.
- View available assignments.
- Upload assignment files.
- Track their assignment submission status.
- Receive email updates about assignment openings, deadlines, and results.

7. Functional Requirements

Functional requirements describe what the system shall do.

7.1 Authentication Requirements

FR-01 – User Login: The system shall allow an authorized Student, Teacher, or Administrator to log in using valid credentials.

FR-02 – Login Validation: The system shall validate the credentials provided by the user.

FR-03 – Invalid Login Handling: The system shall display an appropriate error message when invalid credentials are provided.

FR-04 – User Logout: The system shall allow an authenticated user to log out.

7.2 User Management Requirements

FR-05 – Add Student: The system shall allow an authorized administrator to add a student.

FR-06 – Add Teacher: The system shall allow an authorized administrator to add a teacher.

FR-07 – Manage Users: The system shall allow the administrator to manage registered teachers and students.

7.3 Assignment Management Requirements

FR-08 – Create Assignment: The system shall allow an authorized teacher to create an assignment.

FR-09 – Assignment Details: The system shall store assignment information including relevant title, description, opening time, and deadline information.

FR-10 – Assignment Access: The system shall allow students to view assignments available to them.

FR-11 – Assignment Tracking: The system shall maintain assignment submission status for students.

7.4 Submission Tracking Requirements

FR-12 – Upload Assignment: The system shall allow a student to upload an assignment file.

FR-13 – Record Submission: The system shall record the student's submission against the relevant assignment.

FR-14 – Submission Status: The system shall provide visibility of students who have submitted and students who have not submitted.

FR-15 – Teacher Review: The system shall allow teachers to view submitted assignment files.

7.5 Grading and Reporting Requirements

FR-16 – Assign Marks: The system shall allow teachers to assign marks to submitted assignments.

FR-17 – Record Results: The system shall store assignment marks/results for the relevant student and assignment.

FR-18 – Result Access: The system shall make recorded assignment results available to the relevant student.

7.6 Email Notification Requirements

FR-19 – Assignment Opening Notification: The system shall send automated email notifications to students when an assignment becomes available.

FR-20 – Deadline Notification: The system shall send automated email notifications regarding assignment deadlines.

FR-21 – Result Notification: The system shall send automated email notifications when assignment results are available.

8. Non-Functional Requirements

8.1 Security

NFR-01 – Authentication: Protected system functions shall require user authentication.

NFR-02 – Authorization: System functions shall be restricted according to the user's role.

NFR-03 – Password Security: User passwords shall be stored securely and not as plain text.

8.2 Performance

NFR-04 – Response Time: Commonly used pages and API requests should respond within an acceptable time under normal operating conditions.

NFR-05 – Database Performance: The database shall support efficient storage and retrieval of assignment, submission, user, and grading information.

8.3 Usability

NFR-06 – User Interface: The system shall provide a simple, understandable, and consistent interface.

NFR-07 – Validation Messages: The system shall provide clear messages when input is invalid or an operation cannot be completed.

8.4 Reliability

NFR-08 – Data Integrity: The system shall maintain the integrity and consistency of stored academic and user information.

NFR-09 – Error Handling: The system shall handle expected application errors without unexpected termination.

8.5 Availability

NFR-10 – Availability: The application should be available during the institution's normal operating periods when deployed in the target environment.

8.6 Maintainability

NFR-11 – Maintainable Architecture: The system shall use a modular architecture that allows changes to be implemented with minimal unnecessary impact on existing functionality. **8.7**

Portability

NFR-12 – Browser Compatibility: The web application should operate correctly on commonly used modern web browsers.

8.8 Responsiveness

NFR-13 – Responsive Interface: The user interface should adapt to common desktop, tablet, and mobile screen sizes.

9. User Stories

US-01 – As a registered user, I want to log in using my credentials so that I can access the functions available to me.

US-02 – As an administrator, I want to add students and teachers so that authorized users can access the system.

US-03 – As a teacher, I want to create assignments so that students can complete and submit their work.

US-04 – As a student, I want to view assignments so that I know what work is required.

US-05 – As a student, I want to upload an assignment so that my work can be submitted.

US-06 – As a teacher, I want to see who has submitted and who has not so that I can track completion.

US-07 – As a teacher, I want to grade submissions so that students receive marks.

US-08 – As a student, I want to receive assignment and result emails so that I do not miss important updates.

US-09 – As an authenticated user, I want to log out so that my account remains protected.

10. Acceptance Criteria

AC-01 – Login: Given a registered user is on the login page, when valid credentials are entered and Login is selected, the system shall authenticate the user and display the appropriate page. Invalid credentials shall produce an appropriate error message.

AC-02 – Add User: Given the administrator is authenticated, when valid teacher or student information is entered and saved, the user record shall be stored and a success message shall be displayed.

AC-03 – Create Assignment: Given the teacher is authenticated, when valid assignment information is entered and saved, the assignment shall be created and made available according to its opening and deadline information.

AC-04 – Submit Assignment: Given a student is authenticated and an assignment is available, when a valid assignment file is uploaded, the submission shall be recorded against that assignment.

AC-05 – Submission Tracking: Given a teacher is authenticated, when the teacher opens an assignment, the system shall display submitted and pending students.

AC-06 – Grade Assignment: Given a teacher has access to a submitted assignment, when marks are entered and saved, the result shall be stored and a success message shall be displayed.

AC-07 – Email Notifications: When the relevant assignment event occurs, the system shall send the configured notification to the applicable student.

AC-08 – Logout: Given a user is authenticated, when Logout is selected, the authenticated session/token shall be terminated and the user shall return to the login page.

11. MoSCoW Prioritization

Priority	Requirements
Must Have	Login, Logout, role-based access, assignment creation, student assignment upload, submission tracking, grading, user management, basic security
Should Have	Clear validation, user feedback, submission statistics, automated assignment/result notifications
Could Have	Basic dashboard and additional convenience features where time permits
Won't Have in v1.0	Advanced features not defined in the initial project scope

12. Product Backlog

ID	Product Backlog Item	Priority	Related Requirements
US-01	User Login	Must	FR-01, FR-02, FR-03
US-09	User Logout	Must	FR-04
US-02	Administrator User Management	Must	FR-05, FR-06, FR-07
US-03	Create Assignment	Must	FR-08, FR-09
US-04	View Assignment	Must	FR-10
US-05	Upload Assignment	Must	FR-12, FR-13
US-06	Submission Tracking	Must	FR-14, FR-15
US-07	Grading	Must	FR-16, FR-17
US-08	Email Notifications	Should	FR-19, FR-20, FR-21

13. Use Cases

13.1 Use Case List

Use Case ID	Use Case	Primary Actor
UC-01	Login	Administrator / Teacher / Student
UC-02	Manage Users	Administrator
UC-03	Create Assignment	Teacher
UC-04	View Assignment	Student
UC-05	Submit Assignment	Student
UC-06	Track Submissions	Teacher
UC-07	Grade Assignment	Teacher
UC-08	Receive Notifications	Student
UC-09	Logout	Administrator / Teacher / Student

13.2 Use Case Examples

Use Case ID: UC-02

Name: Manage Users

Actor: Administrator

Preconditions: Administrator must be authenticated

Trigger: Administrator selects user management

Main Flow: Enter user information → Validate → Save user → Display confirmation

Alternative Flow: Invalid or duplicate user information

Postcondition: User record is stored successfully

Use Case ID: UC-03

Name: Create Assignment

Actor: Teacher

Preconditions: Teacher must be authenticated

Trigger: Teacher selects Create Assignment

Main Flow: Enter details → Validate → Save assignment → Publish according to opening time

Alternative Flow: Invalid data

Postcondition: Assignment is stored successfully

Use Case ID: UC-05

Name: Submit Assignment

Actor: Student

Preconditions: Student must be authenticated and assignment must be available

Trigger: Student selects an assignment and uploads a file

Main Flow: Select file → Validate file → Upload → Record submission → Display confirmation

Alternative Flow: Invalid file or submission after deadline

Postcondition: Submission is stored successfully

Use Case ID: UC-07

Name: Grade Assignment

Actor: Teacher

Preconditions: Teacher must be authenticated and submission must exist

Trigger: Teacher opens a submitted assignment

Main Flow: Review submission → Enter marks → Save result → Notify student

Alternative Flow: Invalid marks

Postcondition: Result is stored successfully

14. Business Rules

- BR-01 – Only authenticated users may access protected system functionality.
- BR-02 – Access to system functions shall depend on the user's assigned role.
- BR-03 – Only authorized administrators may add and manage teachers and students.
- BR-04 – Only authorized teachers may create assignments and grade submissions.
- BR-05 – Students may upload submissions only for assignments available to them.
- BR-06 – Each submission shall be associated with the relevant student and assignment.
- BR-07 – Assignment deadlines shall control whether submissions are accepted.
- BR-08 – Recorded marks shall be associated with the relevant assignment and student.
- BR-09 – Student email addresses used for notifications shall follow a valid email format.
- BR-10 – Required information must be provided before a record can be saved.
- BR-11 – Passwords must not be stored in plain text.

15. Data Requirements

Version 1.0 requires the system to maintain core user, assignment, submission, grading, and notification information.

15.1 User Information

- User ID
- Username / login identifier
- Password
- Role
- Email address

15.2 Student Information

- Student identifier / index number
- Name
- Email
- Contact information where required

15.3 Teacher Information

- Teacher / staff identifier
- Name
- Email
- Contact information where required

15.4 Assignment Information

- Assignment ID
- Title
- Description
- Opening time
- Deadline
- Teacher reference

15.5 Submission Information

- Submission ID
- Assignment reference
- Student reference
- Uploaded file
- Submission date/time
- Submission status

15.6 Grading Information

- Result / marks
- Teacher reference
- Grading date/time

15.7 Notification Information

- Notification type
- Recipient
- Assignment reference
- Notification date/time
- Notification status

Detailed database structures, relationships, normalization, constraints, and SQL schemas will be documented separately during the Database Design phase.

16. System Constraints

8. The application will be developed as a web-based system.
9. React will be used for the frontend.
10. FastAPI will be used for the backend.
11. MySQL will be used as the database.
12. Git and GitHub will be used for version control.
13. REST APIs will be used for communication between frontend and backend.
14. Postman will be used for API testing.
15. Version 1.0 will focus on the assignment management workflow and essential user administration.

17. Assumptions

16. The institution will provide authorized users to test the system.
17. Users will have access to a modern web browser.
18. Users will receive appropriate login credentials.
19. Student and assignment information entered into the system will be provided by authorized personnel.
20. The final deployment environment will be defined at a later stage.
21. Email services required for notifications will be available in the target environment.
22. Requirements may evolve after Version 1.0 and will be handled through controlled software changes.

18. Requirements Traceability Matrix

The Requirements Traceability Matrix (RTM) provides traceability between requirements, user stories, acceptance criteria, and future testing activities.

Requirement	User Story	Acceptance Criteria	Test Case
FR-01 Login	US-01	AC-01	TC-01
FR-04 Logout	US-09	AC-08	TC-02
FR-05 Add Student	US-02	AC-02	TC-03
FR-06 Add Teacher	US-02	AC-02	TC-04
FR-08 Create Assignment	US-03	AC-03	TC-05
FR-10 View Assignment	US-04	—	TC-06
FR-12 Upload Assignment	US-05	AC-04	TC-07
FR-14 Submission Tracking	US-06	AC-05	TC-08
FR-16 Assign Marks	US-07	AC-06	TC-09
FR-19–21 Notifications	US-08	AC-07	TC-10

19. Version 1.0 Requirements Baseline

After review and approval, the requirements contained in this document shall become the Requirements Baseline v1.0. The baseline establishes the official reference point for Version 1.0 development.

- New requirements shall not be added directly to the approved Version 1.0 scope.
- Changes shall be assessed for their impact on requirements, design, database, implementation, testing, and documentation.
- Approved baseline and version information shall be updated when an approved change is implemented.

20. Configuration and Version Control

The approved requirements document shall be maintained under Git version control.

Item	Value
Baseline Name	Requirements Baseline v1.0
Version	1.0
Status	Proposed for Review / Approved after Signoff
Git Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____

21. Version History

Version	Date	Description	Author / Status
0.1	_____	Initial requirements draft	Student Project / Draft
0.2	_____	Requirements reviewed and revised	Student Project / Review
1.0	31st August 2026	Requirements baseline established	Student Project / Proposed for Review

22. Approval

By approving this document, the relevant stakeholders confirm that the requirements defined for Version 1.0 represent the agreed initial scope of the Smart Classroom system.

Role	Name	Signature	Date
Client / Business Representative	_____	_____	_____
Project Team Representative	_____	_____	_____
Lecturer / Supervisor	_____	_____	_____

Appendix B – Initial Technology Environment

Component	Technology
Frontend	React
Backend	FastAPI
Database	MySQL
Version Control	Git
Repository	GitHub
API Testing	Postman

Appendix C – Project Lifecycle

The initial project shall follow the following lifecycle:

Project Proposal → Requirements Engineering → Requirements Baseline → System Analysis & Design → Database Design → Backend Development → Frontend Development → Integration → Testing → Version 1.0 Release

After the first release: Change Request → Impact Analysis → Approved Change → Implementation → Testing → New Release

Appendix D – Baseline Statement

Requirements Baseline v1.0

The requirements contained in this document constitute the proposed Version 1.0 requirements baseline for the Smart Classroom system. Any modification after baseline approval must be controlled through the agreed software change management process.

Baseline Tag	requirements-v1.0
Baseline Date	_____
Approved By	_____
Approval Signature	_____

Appendix E – Project Document Sequence

The Requirements Baseline is positioned within the overall practical project as follows: 01

– Initial Project Proposal → 02 – Project Plan / Practical Roadmap → 03 – SRS +

Requirements Baseline v1.0 → System Design → Database Design → Development →
Testing → Version 1.0 Release → Controlled Future Changes